

NANCY KAIM

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EDUCATION

Master of Computer Application Indira Gandhi Delhi Technical University for Women (IGDTUW)	2026 – Present
B.Sc. (Hons.) Research (Computer Science) Indraprastha College for Women, University of Delhi 8.091 CGPA (first division)	2022-2026
Senior Secondary (XII) — CBSE St. John's Public School (80.2%)	2021
Secondary (X) — CBSE St. John's Public School (80%)	2020

RESEARCH EXPERIENCE

Undergraduate Dissertation — University of Delhi 2025 – 2026

Optimisation of Internet of Things Technology in Military Domain | Supervisor: Prof. Archana Singhal

- Designed a severity-aware selective retraining framework for Intrusion Detection System in IoT network, combining label-free drift detection (PSI, JSD), model-driven signals (confidence, entropy), and **SHAP-based explainability** across six classifiers (LR, DT, RF, KNN, XGB, SVM) on the TON_IoT dataset.
- Achieved **89.86% mean accuracy** with only **13 updates** across 8 drift windows, outperforming blind retraining (**81.09%, 48 updates**) by **+8.77%** while cutting retraining frequency by **72.92%**.

Research & Development Intern — WEmpower Oct 2024 – May 2025

- Conducted empirical R&D study on maternal education's impact on child development; contributed to research work and mentored high-school students in research fundamentals, including academic paper analysis, literature review, and research synthesis.

Research & Innovation Member — Girl Up Rooh (UN Foundation) Oct 2023 – Oct 2024

- Produced research-driven content on gender equality and youth empowerment for national magazines; analysed societal data for strategic storytelling.

PUBLICATIONS

[1] Verma, T., Rajak, M., **Kaim, N.**, Sharma, N., Nisha, & Singh, H. P. (2025, December). Bridging Gaps in IoT Literature: A Thematic Categorization of Challenges and Resolutions. In *International Conference on Information and Communication Technology for Competitive Strategies* (pp. 197-209). Cham: Springer Nature Switzerland.

[2] Rajak, M., **Kaim, N.**, Sharma, N., Verma, T., Singh, H. P., & Nisha. (2026). The Evolving Paradigm of IoT: A Systematic Literature Survey of Challenges, Solutions, and Emerging Technologies. *Blockchain Solutions for Securing Internet of Things Networks*, 1-37. (Wiley)

AWARDS & RECOGNITIONS

- **Best Paper Award** “Optimisation of Internet of Things Technology in Military Domain ” — **IoT Track, ICET-AICS 2026 International Conference.**
- Paper Poster Presentation (UG dissertation) — Undergraduate Research and Entrepreneurship Exhibition 2026, University of Delhi (**selected by Indraprastha College for Women Research Committee**).
- **Qualified GATE 2026** in two disciplines: Data Science & AI (82.0 Percentile) and Computer Science (70.6 Percentile).

TECHNICAL PROJECTS

- **Adaptive Intrusion Detection (Dissertation Project 2026):** Python, Scikit-learn, SHAP, Pandas | Built and evaluated 6 ML models for network intrusion detection under simulated concept drift (TON_IoT dataset); designed label-free model drift/adaptability analysis; used SHAP value stability to audit model explainability and decision consistency under drift; engineered pre-processing pipelines (scaling, one-hot encoding) serialized via Pickle for real-time, window-based inference simulation.
- **LLM AI Chatbot for Judicial Intelligence (2024):** Fine-tuned GPT-2 to deliver judiciary information (case status, eCourt services, NJDG integration) with scalable, multilingual-ready architecture.
- **LLM Data Retrieval Chatbot (Woodpecker Analytics Hackathon, 2024):** Deployed Google TAPAS model via Python/Streamlit for natural-language SQL-free querying over structured datasets.
- **Smart Plant Pot (2024) :** Arduino-based soil-moisture monitoring system with real-time cloud data visualisation and a JavaScript/HTML interface.

SKILLS

- **Languages & ML:** Python, Transformers, scikit-learn, TensorFlow, XAI (SHAP), Streamlit, Pandas, NumPy, Matplotlib
- **Research Methods:** Concept Drift Detection, Explainable AI (SHAP), Systematic Literature Review, Statistical Analysis (Spearman, Cosine Similarity)